# Comprehensive Design & UI/UX Architectural Specification of **pages/index.vue**

System: CowCare AI (Heat Stress Intelligence for Dairy Farms)
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## 1. Executive Overview & Design Intent

The public-facing `pages/index.vue` serves as the primary digital gateway for CowCare AI. It is architected not merely as a marketing landing page, but as an Enterprise-Grade Agritech Product Showcase & Interactive Sandbox.

Core Objectives:

1. Evidence-Based Storytelling: Shift the narrative of tropical smallholder dairy management from "Reactive" (waiting for milk drop after 5 days) to "Proactive" (3–7 day AI early warnings).
2. Interactive Scientific Verification: Provide farmers and agritech evaluators with a real-time sandbox that models the National Research Council (NRC, 1971) heat stress equations using Thai-Holstein biological thresholds (THI $76.0$).
3. Zero Administrative Exposure: Ensure public users have access to educational, simulator, and general member login flows (`/login`), while strictly segregating administrative management surfaces (`/Admin`) behind rigorous authentication guards.

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## 2. Visual Identity & Design Tokens (Biophilic Agritech System)

The visual design is grounded in the Biophilic Agritech Design System, balancing biological warmth with precision industrial engineering.

Color Palette & Semantic Tokens

| Design Token | Hex Code | Semantic Role | Psychological / Functional Impact |
|---|---|---|---|
| Warm Ivory Canvas | `#FFFDF7` | Global Page Background | Replaces stark `#FFFFFF` glare with an organic, tactile tone that reduces eye strain in outdoor agricultural environments. |
| Deep Forest Green | `#24433A` / `#163026` | Primary Brand & Structure | Represents ecological balance, vitality, and enterprise reliability. |
| Deep Night Forest | `#19362E` | High-Tech Section Accent | Inverted background for Section 4 (Predictive AI) to signal computational depth. |
| Terracotta Amber | `#D87939` / `#F2BD7C` | Primary Action / Accent | Evokes thermal energy, critical inflection points, and predictive chart lines. |
| Comfort Zone | `#10B981` (Bg `#ECFDF5`) | THI $< 76.0$ | Optimal bovine thermal comfort, standard milk yield, and zero heat load. |

| Design Token | Hex Code | Semantic Role | Psychological / Functional Impact |
|---|---|---|---|
| Mild Stress | `#F59E0B` (Bg `#FFFBEB`) | THI $76.0 - 79.9$ | Critical onset threshold; initiation of milk drop ($-0.28$ kg/THI). |
| Moderate Stress | `#EA580C` (Bg `#FFF7ED`) | THI $80.0 - 83.9$ | Elevated somatic cell count; doubling of milk loss ($-0.56$ kg/day). |
| Severe Heat Stress | `#E11D48` (Bg `#FFF1F2`) | THI $\geq 84.0$ | Emergency heat load; risk of embryonic death and acidosis (FPR $< 1.2$). |

Typographic System

- Type Family: `'Noto Sans Thai', 'Inter', -apple-system, sans-serif` for high legibility across bilingual Thai/English scientific copy.
- Tabular Numerals (**`tabular-nums`**): Enforced across all sensor readouts, sliders, and telemetry counters to prevent jitter and layout shifts during dynamic calculation.
- Fluid Type Scaling: Headings leverage CSS `clamp()` (e.g., `clamp(2rem, 5vw, 3.4rem)`) for continuous responsive scaling across mobile, tablet, and widescreen monitors.

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## 3. Structural Decomposition of the 7 Key Sections

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┌──────────────────────────────────────────────────────────────────┐
│ 1. Hero Section & Interactive Sandbox (Split Grid 7 : 5 Columns) │
│    ├── Value Proposition, Eyebrow & Public Action CTAs (/login)  │
│    └── Real-time NRC 1971 Calculator, 4 Thai Presets, Continuous Gauge │
├──────────────────────────────────────────────────────────────────┤
│ 2. Scientific Signal Strip (Deep Green Banner #24433A)           │
│    ├── THI 76.0 (Onset)          ├── 5 Days (Lag Effect)         │
│    ├── 14 Days (Lookback)        └── 3–7 Days (Proactive Horizon)│
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│ 3. 3-Layer Architecture & Bento Feature Matrix (Sense → Rec → Cap)│
│    ├── Card 01: SENSE (ESP32-S3 + SHT40 + YOLO26m Edge Vision Mockup) │
│    ├── Card 02: RECOMMEND (LSTM Time-Series + Gemini LINE Advisory)   │
│    └── Card 03: CAPITALIZE (Digital Logbook + ESG Green Credit)       │
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│ 4. Predictive Analytics & LSTM Brain Engine (Dark Night Forest #19362E)│
│    ├── Node-Based Architecture Flow: Inputs → LSTM Brain → Outputs    │
│    └── Comparative Bento: Reactive (Traditional) vs Proactive (CowCare)│
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│ 5. Digital Logbook & ESG Preview (Bridging the Digital Divide)   │
│    ├── Thai Smallholder Barn Realities & 17.5% Methane Reduction │
│    └── Mockup Digital Logbook Card with 7-Day Dynamic SVG Sparkline │
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│ 6. Academic Foundation & Peer-Reviewed Citation Matrix (2022–2026)│
│    ├── Boonkum et al. (2024)     ├── Hasan et al. (2026)         │
│    ├── Onan-Martinez (2026)      └── Mulajkar & Abdourhamane (2026)│
├──────────────────────────────────────────────────────────────────┤
│ 7. Final Conversion CTA & Global Footer                          │
│    ├── Public Sign-in Link (/login) & Architecture Anchor (#solution) │
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└── Global Semantic <AppFooter />
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## 4. Deep-Dive: Engineering & Interaction Architecture

### 4.1 Section 1: Hero Section & Interactive Telemetry Sandbox

- Left Column (7 Cols — Value Narrative):
  - Pulsing Eyebrow Badge: `"PRECISION LIVESTOCK FARMING • MULTIMODAL AI"` indicating live technology status.
  - Problem Statement Callout: Highlights the urgency that Thai crossbred cows experience THI $\geq 76.0$ for 250 days per year (67% of the annual cycle).
  - Public Action CTA: Contains a primary button redirecting to member authentication (`/login`) and a secondary smooth-scroll button focusing the simulator.
- Right Column (5 Cols — Interactive Telemetry Sandbox):
  - Real-time NRC Formulation: Executes reactive client-side computation:

$$\text{THI} = (1.8 \times T + 32) - (0.55 - 0.0055 \times \text{RH}) \times (1.8 \times T - 26)$$

  - 4 Quick Presets for Thai Regional Scenarios:
    1. Chiang Mai (Winter): $24.0^{\circ}$C, $55\%$ RH $\rightarrow$ THI $70.8$ (Comfort)
    2. Saraburi (Summer): $34.0^{\circ}$C, $60\%$ RH $\rightarrow$ THI $83.8$ (Moderate)
    3. Pak Chong / Korat (Rainy Season): $31.5^{\circ}$C, $82\%$ RH $\rightarrow$ THI $85.2$ (Severe)
    4. Heatwave Crisis: $36.5^{\circ}$C, $78\%$ RH $\rightarrow$ THI $92.5$ (Extreme Severe)
  - Continuous Color Gauge Bar: A 4-color gradient bar with a smoothly positioned pointer (`transition: left 0.25s`) mapping the current value across a $60 - 95+$ scale.
  - Immediate Impact & Advisory Output: Calculates live milk loss rates (e.g. $-0.28$ to $-0.42$ kg/THI), subclinical acidosis risk (FPR $< 1.2$), and enteric methane spikes ($+17.5\%$), paired with immediate mitigation instructions (fans, mist sprayers, Sodium Bicarbonate buffers).

### 4.2 Section 2: Scientific Signal Strip

- Anchors the platform's credibility with 4 responsive metrics:
  - **`THI 76.0`**: Physiological onset threshold in tropical crossbred cattle.
  - **`5 Days`**: Biological lag effect before milk production bottoms out.
  - **`14 Days`**: Historical lookback window for thermal accumulation analysis.
  - **`3–7 Days`**: Lead time horizon for proactive intervention.

### 4.3 Section 3: 3-Layer Architecture & Bento Matrix

- Sense (Card 01 — Deep Forest **`#24433A`**):
  - Details the industrial hardware spec: ESP32-S3 (N16R8) + SHT40 sensor over RS485 Modbus RTU.
  - YOLO26m Edge Vision Viewfinder: A simulated real-time camera interface running the project's actual fine-tuned `YOLO26m` model (`best1.pt`), displaying the 4 core bovine behavior classes:
    - `standing` (94.2% confidence)
    - `lying down` (91.8% confidence)
    - `drinking` (89.4% confidence)
    - `feeding` (88.5% confidence)
- Recommend (Card 02 — Glass Card):
  - Highlights LSTM time-series modeling coupled with Context-Aware Gemini GenAI translating telemetry into farmer-friendly Thai alerts via LINE Messaging API.
- Capitalize (Card 03 — Glass Card):
  - Bridges technical data to financial incentives: Digital Logbooks on PostgreSQL / Supabase, ESG compliance, and parametric weather index insurance.

4.4 Section 4: Predictive Analytics & LSTM Brain Engine

- Inverted Dark Mode (**#19362E**): Distinct aesthetic shift for deep-learning concepts.
- Node-Based Model Flow Diagram:
  - Inputs: SHT40 Microclimate Signals + 14-Day Accumulation + YOLO26m Behavioral Vectors.
  - Brain Core: LSTM Neural Network displaying 96.08% Validation Accuracy.
  - Outputs: 3–7 Day THI Trajectory, Milk Loss Prevention, and LINE Broadcasts.
- Comparative Analysis Matrix: Side-by-side comparison illustrating why traditional reactive responses fail due to the 5-day lag effect, contrasted with CowCare AI's proactive preemptive cooling.

4.5 Section 5: Digital Logbook & ESG Preview

- Digital Divide Context: Explains that 80%+ of Thai smallholders operate low 2.7m barns with heat-radiating zinc roofs and lack the capital for expensive wearable sensors.
- Interactive Mockup Logbook Card:
  - Weekly Average THI ($74.8$), Cumulative Heat Hours ($12.5$ hrs), and Transmission Completeness ($98.4\%$).
  - Custom 7-Day SVG Sparkline: Vector-drawn line graph plotting weekly thermal trajectory with an active dashed reference line at the critical THI $76.0$ mark and a highlighted Friday peak.

4.6 Section 6: Academic Foundation & Peer-Reviewed Citations

- Bento grid quoting 4 international peer-reviewed papers (2022–2026):
  1. Boonkum et al. (2024, Animals): Thai-Holstein THI 76 threshold and milk/conception rate degradation.
  2. Hasan et al. (2026, Comput. Electron. Agric.): Discovery of the 5-day lag effect across 82,000+ cattle.
  3. Onan-Martinez et al. (2026, J. Dairy Sci.): Documentation of $+17.5\%$ methane emission intensity under thermal stress.
  4. Mulajkar et al. & Abdourhamane (2026): Edge AI fusion and 14-day LSTM forecasting performance.

4.7 Section 7: Final Conversion CTA & Footer

- A clean, cohesive closing block driving visitors to `/login` or internal documentation anchors (`#solution`), completed with `<AppFooter />`.

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## 5. Security & Access Segregation Architecture

In response to strict security and authorization requirements: 1. Public Surface Neutrality: - All direct hyperlinks and mentions of `/Admin` ("Command Center") have been completely removed from the public landing page. - Any external visitor seeking system entry is directed strictly to `/login`. 2. Layout Route Guard (**`layouts/admin.vue`**): - Enforces an automated client-side check on `onMounted`: - If `localStorage.getItem('token')` is missing $\rightarrow$ immediate redirection to `/login`. - If the authenticated user's role is not `'admin'` (e.g. general member, farmer) $\rightarrow$ access is rejected and redirected to `/login`. 3. Foundation for Upcoming User Portal: - Prepares the groundwork for the upcoming User Interface (`/user`), ensuring that `/login` serves as the centralized, role-based traffic controller.

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## 6. Performance & Motion Engineering

- Native IntersectionObserver: Replaces heavy third-party scroll libraries. Scans `.reveal` elements with a threshold of `0.12` and activates the `.is-visible` state.
- Hardware Acceleration: All transition animations utilize `translate3d(0, 0, 0)` with `will-change: opacity, transform` to ensure smooth 60fps rendering on mobile devices.
- Accessibility (**`prefers-reduced-motion`**): Includes a global media query neutralizing animation durations to `0.01ms` for users with vestibular sensitivities.

- History Scroll Restoration: Sets `history.scrollRestoration = 'manual'` and forces `window.scrollTo(0, 0)` on load to guarantee predictable layout rendering upon browser refreshes.